Solutions to Review Questions for Lecture 7

2025-02-13

1. The blanks are the scatter plot and spread. An example that is not football shaped would look like a parallelogram where the points are evenly spread in each vertical strip.

2a. The value $x = 140$ is 2 SDs above the mean and the correlation is 0.8. We would predict that y would be $2 * 0.8 = 1.6$ SDs above the mean, which would be $y = 200 + 1.6 * 20 = 232$.

2b. We would estimate the SD in a vertical strip as the rms error, or $\sqrt{1 - 0.8^2} * 20 = 12$.

2c. It is not possible to say since we do not know that the points inside a vertical strip are approximately normally distributed.

2d. The value 200 is $(200 - 232)/12 = -2.67$ rms errors below the mean in the vertical strip, while the value 264 is $(264 - 232)/12 = 2.67$ rms errors above the mean. The largest possible fraction of values outside the range would be $1/2.67^2 = 14\%$ by the Chebyshev inequality.

3a. The dose is randomized, the subjects and doctors are blinded, and we can think of the control group as the subjects with 0 dose. We can think of this as a randomized controlled double-blind experiment.

3b. This is false because the trend is not linear. Years lived increases as dose increases when the dose is less than 2 mg.

3c. This is false because the years lived drops below the years lived at 0 mg for larger doses.

3d. This is likely true based on the portion of the scatter plot corresponding to higher doses.

3e. This is false since the formula assumes a linear trend which does not hold for this data.

3f. This is false since the data does not follow a linear trend.

4.

```
MAXABS = function(x) {
  max = abs(x[1])
  for (i in (2:length(x))) {
    if (abs(x[i]>max))
    {   max = abs(x[i])
    }
  }
  return(max)
}
MAXABS(c(2,9,5,-4,0))
```

```
## [1] 9
```